

HOW WE TEST SMARTPHONES

Now that you've finally bought the smartphone that you wanted, you can take more time to test it out. The first and foremost thing to do is to establish the testing conditions. There have to be constants, so that each phone that you test can be compared to others. For cameras, you must use the same subjects and conditions, while performance must be gauged based on precise and identical tests between phones. Here's how we do it.

Display: For our tests, we first turn the display up to full brightness. The brightness levels may differ, but it ensures that each phone is as bright as it can be for the tests.

We start with the simple tests, similar to what you read hear for testing in a retail store. The difference is that the phone is always kept under the same light and at the exact location, so that the surrounding conditions are also exactly the same for each phone. We check for colour shifts and gauge the viewing angles, which gives us a good idea of what we're working with.



Next, we take the phone into a completely dark room and open various images on the display, examining the colours and brightness. This is not a real world test, but one that gives us a good idea of what the display is capable of without any outside interference.

For a more objective look, we use the lagom.nl tests on the browser to get an even better idea of the display. This is then followed by apps like AnTuTu, which offer a display test of their own. This really gives you an idea of the capacity of a display. This is done in a more real world settings, but in an area where the light is always the same.

Lastly, we take the best performing smartphones from the price segment that the phone belongs to and examine the display next to these phones. We use the same images on all the phones and examine the difference.

In some cases, like in Super AMOLED and AMOLED displays, there's a visible yellow-ish tinge on the screen, which is visible irrespective of what the light conditions may be.

Lastly, we take the display out in the sunlight and examine the visibility outdoors. This step is very important.

Performance

Processor: As mentioned before, the processor is one of the most important aspects of a smartphone, so we spend the most amount of time testing it.

While most use benchmarks alone, here at Digit, we believe the real world performance is not reflected by benchmarks. Benchmarks are good for gaining an objective perspective on the performance and one that can be easily compared to other smartphones. To start with, we use the AnTuTu, Geekbench 3, PC Mark for Android and 3D Mark benchmarks to test the processor's performance. This is always done on a freshly set up device and each benchmark is run from a fresh reboot. In order to ensure that the scores you're getting are as accurate as possible, it is advisable to run each benchmark multiple times, and always from a fresh reboot.

You may be wondering why 3D Mark is being used as a processor test, well, it is not exactly used that way. The 3D Mark benchmark is a graphics test, but one that takes into account the entire system as a whole. Hence, it is a very good benchmark for testing SoCs on smartphones. The AnTuTu and Geekbench tests are the major indicators of the processor performance though.

The Geekbench 3 test gives you comparative scores for single core and multi-core performance. This can be directly translated to your apps, some of which require only one CPU core to be running, while others need multiple. So, a phone that shows good single core performance but only satisfactory multi-core performance, can be expected to slack off on more intensive applications and tasks.

The AnTuTu score breaks this down further, with scores for Integer and Floating Point performance. Of course, the Floating Point values are always lower than Integer values, but you wouldn't want them to be very low. Integers are easier to compute, which is why the values are higher.

The AnTuTu test also gives another very important number, which many overlook, but more on that later. The benchmark tests are run once again after the phone has been loaded with apps and used for a while. You will almost always notice a drop in the scores here, which is expected, but the drop shouldn't be very high. For example, a Moto Turbo shows about a 2% drop after it has been loaded with apps like any regular phone would be. It's a very good way to get an objective idea of the real world performance of the phone.

GPU: Have you seen mobile games today? They're becoming better by the day, especially with companies like Marvel and Warner Brothers taking interest. Our principle GPU test is GFX Bench, which is known for its GPU tests. In fact, the Manhattan tests on GFX Bench were added quite recently, in order to provide for the newer GPUs coming into the market nowadays. It's the perfect benchmark to use for GPU performance comparison.

While GFX Bench provides a lot of tests, the most important are the T-Rex and Manhattan tests, both onscreen and offscreen. On the onscreen tests, the benchmark simulates a gaming environment, taking into account the smartphone's display as well. The offscreen tests on the other hand are done without taking the display into account. Naturally, the scores for offscreen are always higher than onscreen. This is because 720p, 1080p or 2k displays are not easy on a processor.

Heat: A question we constantly get about almost every smartphone today is whether it heats up. Well, every phone does, especially if you live in the Indian summers. That said, some heating is fine, absurd or abnormal heating is not. The GFX Bench and 3D Mark tests come in very